CHAPTER 1: THE PROJECT AND ITS BACKGROUND
Chapter 1 establishes the **foundation of the study**. Based on the attached document, this chapter must:

- Introduce the **problem situation**
- Identify the **stakeholders or clients**
- Present the **proposed project or solution**
- State **clear objectives**
- Define the **scope and limitations**
- Identify **design or research constraints**
- Cite **applicable standards or guidelines**
- Explain the **design or research process followed**

The chapter moves from **context** → **justification** → **solution** → **structure** → **methodology**

Chapter Opening Paragraph
What to Write
A short introductory paragraph explaining:

- That the chapter presents the **background of the project/study**
- That it discusses the **problem, client, objectives, constraints, standards, and process**

What to Include

- A general statement describing the role of Chapter 1
- A list (in sentence form) of the major subtopics covered

This paragraph acts as a **roadmap** for the reader

1.1 The Problem
To clearly describe the **problem that motivated the study** and justify why it needs to be addressed.
What to Write

1. **Background of the subject area**
    - Describe the system, process, or phenomenon being studied
    - Provide historical or contextual information
2. **Current practice or situation**
    - Explain how things are currently done
3. **Issues or limitations**
    - Identify inefficiencies, risks, gaps, or challenges
4. **Evidence of the problem**
    - Cite studies, statistics, observations, or expert findings
5. **Existing solutions**
    - Briefly describe current solutions
    - Clearly explain their limitations or drawbacks
6. **Need for improvement**
    - State why a better or alternative solution is necessary
7. **Broader relevance**
    - Mention contributions to productivity, safety, efficiency, quality of life, or development goals

What to Include

- Research citations
- Logical progression from general to specific
- A concluding paragraph that **clearly states the problem gap**

1.2 The Client
To identify **who experiences the problem** and validate it through stakeholders.
What to Write

1. **Identification of the client or users**

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    - Individuals, professionals, organizations, or communities
2. **Description of their current practices**
    - How they interact with the problem
3. **Observed difficulties**
    - Risks, inefficiencies, or discomfort experienced
4. **Validation of the problem**
    - Interviews, surveys, consultations, or documented practices
5. **Client needs and expectations**
    - Functional, physical, economic, or usability needs

What to Include

- Descriptive evidence (interviews, observations)
- A clear transition from **user needs to system requirements**
- A summary table converting:
    - Client requirements → Technical or research requirements

1.3 The Project / Solution

To introduce the **proposed solution or study approach** at a conceptual level.

What to Write

1. **General description of the project**
    - What the project/system/study is
2. **Purpose of the solution**
    - How it addresses the identified problem
3. **Key features or components**
    - Major functional parts (high-level only)
4. **Basic operation**
    - Input → process → output description
5. **Expected benefits**
    - Efficiency, accuracy, safety, usability, or effectiveness

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What to Include

- Conceptual explanation only
- No detailed design, algorithms, or results
- Clear link back to the problem and client needs

1.4 The Project Objectives

To state **what the study aims to achieve**.

What to Write

1. **General Objective**
    - One comprehensive statement of the main goal
2. **Specific Objectives**
    - A list of measurable and actionable objectives
    - Each objective should describe a task or outcome

What to Include

- Objectives that are:
    - Clear
    - Achievable
    - Testable
- Alignment with:
    - Problem
    - Client needs
    - Constraints and standards

1.5 Scope and Delimitation
To define **boundaries of the study**.
What to Write

1. **Scope**
   - What the study includes
   - Conditions under which it operates
2. **Delimitations**
   - What the study does not cover
3. **Operational limitations**
   - Materials, participants, environment, or capacity limits

What to Include

- Technical, methodological, or operational limits
- Clear exclusions to prevent misinterpretation
- Honest and explicit statements

1.6 Design Constraints
To explain **factors that limit or influence design or research decisions**.
What to Write

1. **Primary constraints**
   - Economic
   - Environmental
   - Risk
   - Safety
   - Sustainability
2. **Definition of each constraint**
   - What it means
   - How it is measured
3. **Evaluation logic**
   - How constraints affect decision-making
4. **Secondary constraints**
   - Public health, welfare, social, global, cultural (if applicable)

What to Include

- Clear definitions
- Criteria for determining the "best" option
- Explanation of relevance to the study

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1.7 Engineering / Research Standards
To show that the study follows **recognized standards or guidelines**.
What to Write

1. **List of applicable standards**
   - Industry, professional, technical, or ethical standards
2. **Explanation of each standard**
   - What it governs
   - Why it is relevant to the study
3. **Application**
   - How the standard guides design, safety, accuracy, or usability

What to Include

- Official names and versions of standards
- Brief explanations (not full reproductions)
- Reference to appendices if needed

1.8 Engineering / Research Design Process
To explain **how the study was systematically developed**.
What to Write

1. **Overview of the process**
    - Description of the design or research framework
2. **Step-by-step explanation**
    - Identifying the problem
    - Researching the problem
    - Developing solutions
    - Selecting the best option
    - Building or implementing
    - Testing and evaluation
    - Improvement or iteration

What to Include

- Logical progression of steps
- Eng'g Design process diagram
- Connection to the study's context
- Emphasis on iteration and validation

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CHAPTER 2: PROJECT DESIGN

This guide explains **how to write Chapter 2 (Project Design)**.

Chapter 2 explains **how the problem identified in Chapter 1 will be solved**. It focuses on:

- The **alternative designs** considered
- The **final system designs**
- The **hardware and system architecture**
- **Standards and constraints** used to evaluate each design

This chapter is **technical and descriptive**, not theoretical or statistical.

Begin with a short introductory paragraph explaining:

- What this chapter contains
- That multiple designs are explored
- That the designs are based on requirements and constraints defined in Chapter 1

2.1 Discussion of Alternative Designs

What to Write:

This section introduces **all design options** considered for solving the problem.

You must include:

- A brief description of the **overall system concept**
- The **number of alternative designs** explored (e.g., three designs)
- The **main differentiating factor** between the designs (e.g., type of mechanism, control method, technology used)
- A statement that all designs follow:
    - Client or user requirements
    - Safety and engineering standards
    - Constraints discussed in Chapter 1

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Key Points to Include:

- What the system is intended to do (general, not detailed)
- Why multiple designs were necessary
- General material, safety, and sustainability considerations
- List of **standards** applicable to the project (safety, usability, accuracy, etc.)

Do **not** evaluate yet. Only describe and introduce.

2.2 Design 1: [Name of Design]

Each design must follow the **same internal structure**.

2.2.1 Design Description

What to Write:

Explain **what the design is and how it works** at a conceptual level.

Include:

- The main technology or mechanism used
- A definition of the core component(s)
- Why this design is suitable for the application
- General advantages relevant to the project

Avoid:

- Cost comparison
- Final justification (save for later chapters)

2.2.2 Hardware Design

What to Write:

Describe the **physical and mechanical/software/topology structure** of the system.

Include:

- Description of the system layout or assembly
- Degrees of freedom or movement (if applicable)
- Role of each major mechanical component
- Reference to figures or 3D renders
- Explanation of how components are positioned and connected

If applicable, mention:

- Use of CAD or modeling software or any modern software tools
- How components support system functionality

2.2.3 Schematic Design

What to Write:

Explain **how components are electrically or logically connected**.

Include:

- Purpose of schematic diagrams
- Software used to create schematics
- Identification of system blocks (e.g., controllers, power, sensors, actuators)
- Description of how signals flow between components
- Role of main controller(s)

This section is **connection-focused**, not mechanical.

2.2.4 Illustrative Design

What to Write:

Explain **how the entire system works as a process**.

Structure it as (table or matrix):

- **Input** – what activates or provides data to the system
- **Process** – components and algorithms that handle data and control
- **Output** – final action or result of the system

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Include:

- Step-by-step functional explanation
- Interaction between user and system
- Role of displays, sensors, and actuators

2.2.5 Design Standards

What to Write:

List and explain the **engineering, safety, usability, and coding standards** followed.

Include:

- Name and code of each standard
- Why each standard applies to the design
- What aspect it ensures (safety, accuracy, usability, reliability)

2.2.6 Design Constraints

This section evaluates the design using **fixed criteria**. All designs must use the **same constraints**.

Required Constraints:

2.2.6.1 Economics (Material Cost)

- Define economic constraint
- Provide a cost table
- State total cost

2.2.6.2 Safety (Safety Evaluation Score)

- Explain safety evaluation method
- Identify evaluation criteria
- Present a safety scoring table

- State total safety score

2.2.6.3 Risk (Failure Rate)

- Define failure rate
- Present formulas used
- Provide component lifespan table
- Compute and explain failure rate

2.2.6.4 Environmental (Material Consumption)

- Explain material usage evaluation
- Identify materials measured
- Provide material consumption table
- State total material usage

2.2.6.5 Sustainability (Power Consumption)

- Define power consumption constraint
- Present formulas used
- Provide power consumption table
- Compute daily energy use

All formulas, tables, and explanations must be clearly labeled.

2.3 Design 2: [Name of Design]

Repeat **exactly the same structure** used in Design 1:

- Design Description
- Hardware Design
- Schematic Design
- Illustrative Design
- Design Standards
- Design Constraints (with all sub-sections)

Only the **design content changes**, not the structure.

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2.4 Design 3: [Name of Design]

Again, follow the **same format and level of detail** as Designs 1 and 2.
Consistency across designs is critical for later comparison.

Writing Style Requirements

- Formal and technical tone
- Third-person perspective
- Past or present tense consistently
- Clear figure and table references
- No opinions or conclusions

2.5 Software Design

This section explains **how the software component of the system is designed** based on the project requirements.
It focuses on transforming system requirements into a structured software solution that users can interact with.
What to Write:
You must introduce software design as a **crucial development process** that:

- Converts project requirements into software structure
- Defines methods, functions, and system behavior
- Ensures user requirements are satisfied

State that the software design consists of:

- System Algorithm
- Data Flow Diagram
- System Flowchart

- Software User Interface Design

Do **not** include source code. Focus on **design and logic only**.

2.5.1 System Algorithm

To describe the **logical sequence of operations** performed by the system.

What to Write:

- Define the system algorithm as a **step-by-step process**
- Explain that it shows how the system automates or improves a manual process
- Divide the algorithm into the following four required parts:
  - Initialization
  - Input
  - Process
  - Output

What to Include:

- A paragraph explaining each part of the algorithm
- A table summarizing the algorithm with four columns:
  - Initialization
  - Input
  - Process
  - Output
- General descriptions of actions (avoid device-specific names)

The algorithm explanation must clearly show **how inputs are transformed into outputs**.

2.5.2 Data Flow Diagram

To graphically illustrate **how data moves within the system**.

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What to Write:

- Define what a Data Flow Diagram (DFD) is
- Explain its importance in visualizing system processes
- Mention that standard symbols are used to represent:
  - Processes
  - Data inputs
  - Data outputs
  - Data storage

What to Include:

- Description of the overall data flow from user/system input to output
- Step-by-step narrative explanation of the diagram
- Reference to the figure showing the DFD
- Explanation of looping or continuous processes, if any

Avoid interpreting results—only explain the flow.

2.5.3 System Flowchart

To present the **decision-making and control flow** of the software.

What to Write:

- Define the system flowchart
- Explain that it shows the logical progression of system operations
- Describe system behavior from startup to shutdown

What to Include:

- Description of:
  - System initialization
  - User authentication (if applicable)
  - Main interface navigation

    - Conditional decisions (yes/no, true/false)
    - System responses to user actions
- Reference to the flowchart figure

The explanation should follow the flowchart **top to bottom**.

2.5.4 Software User Interface Design

To describe **how users interact with the software visually and functionally**.

What to Write:

- Define user interface design
- Explain the platform used (e.g., touchscreen, desktop, mobile)
- Mention the programming language or framework (general, not code)
- State compliance with relevant software design standards

What to Include:

Describe each interface window or screen, such as:

- Instruction or help screen
- Main menu or dashboard
- Settings or configuration window
- Information or about page
- Error or alert messages
- Login or authentication screens (if applicable)

For each screen:

- State its purpose
- Describe available buttons or controls
- Explain user actions and system responses
- Reference corresponding figures

2.5.5 Other tools or software used (If Applicable)

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CHAPTER 3: DESIGN TRADEOFFS

Chapter 3 explains **how the final design or solution is selected** from several alternatives by analyzing **design tradeoffs**. It demonstrates objective decision-making using:

- Defined constraints
- Quantitative normalization
- Weighted scoring
- Sensitivity analysis

This chapter is **analytical and comparative**, not descriptive.

Chapter Introduction

What to Write:

- Define what a **design trade-off** is
- Explain that improving one characteristic may reduce another
- State that constraints from Chapter 1 and data from Chapter 2 are used
- Clarify that the goal is to identify the **design that best satisfies client requirements**

3.1 Summary of Constraints

To present all constraints used as the **basis for comparison**.

What to Write:

- Define what a constraint is in the context of the study
- Explain why constraints are critical to project success
- State that all alternative designs are evaluated using the same constraints

What to Include:

- A table summarizing all constraints for each design
- The constraints must be identical to those defined in Chapter 1 and measured in Chapter 2
- Typical constraints include:
    - Economic (e.g., cost)
    - Safety
    - Risk or reliability
    - Environmental impact
    - Sustainability

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Notes:

- Briefly interpret the table by stating which design performs best under each constraint
- Do not declare the final (design) winner yet

3.2 Trade-offs

To compare designs by **balancing competing constraints**.

What to Write:

- Define trade-offs as the act of prioritizing certain constraints over others
- State that each design is evaluated against all constraints
- Explain that normalization is required to fairly compare different units

Normalization of Constraints

What to Include:

- Introduce two equations:
    - **Maximization equation** (used when higher values are better)
    - **Minimization equation** (used when lower values are better)
- Clearly define all variables used in the equations:
    - Normalized value
    - Raw value
    - Maximum raw value
    - Minimum raw value

FORMULAS USED IN DESIGN TRADEOFF ANALYSIS (use this as reference, I repeat as reference only)

1. Raw Constraint Values
For each design Di and each constraint Cj, you must first identify the **raw measured value**:

$$X_{ij}$$

Where:

- X_{ij} = raw value of constraint j for design i

Examples:

- Cost in currency units
- Safety score
- Failure rate
- Power consumption
- Material usage

These raw values come **directly from Chapter 2**.

2. Identification of Optimization Type
Before normalization, **each constraint must be classified** as one of the following:
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Minimization Constraints (Lower is Better)
Examples:

- Cost
- Power consumption
- Failure rate
- Material usage

Maximization Constraints (Higher is Better)
Examples:

- Safety score
- Efficiency score
- Accuracy
- Reliability score

This classification determines which normalization formula is used.

3. Normalization of Constraints
Normalization converts values with **different units** into **dimensionless values between 0 and 1**, allowing fair comparison.

3.1 Normalization for Minimization Constraints
Use this formula **when lower values are better**:

$$N_{ij} = \frac{X_{j,\max} - X_{ij}}{X_{j,\max} - X_{j,\min}}$$

Where:

- N_{ij} = normalized value of constraint j for design i
- X_{ij} = raw value of constraint j for design i
- $X_{j,\max}$ = highest raw value among all designs for constraint j
- $X_{j,\min}$ = lowest raw value among all designs for constraint j

Interpretation:

- Best (lowest raw value) - normalized value close to **1**
- Worst (highest raw value) - normalized value close to **0**

3.2 Normalization for Maximization Constraints
Use this formula **when higher values are better**:

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$$N_{ij} = \frac{X_{ij} - X_{j,\min}}{X_{j,\max} - X_{j,\min}}$$

Where all variables retain the same definitions.
Interpretation:

- Best (highest raw value) - normalized value close to **1**
- Worst (lowest raw value) - normalized value close to **0**

4. Importance Weight Assignment
Each constraint is assigned an **importance score** based on client or project priorities.
4.1 Importance Score

$$I_j \in [0, 10]$$

Where:

- I_j = importance score of constraint j

4.2 Conversion to Weight Percentage
Each importance score is converted into a **weight percentage**:

$$W_j = \frac{I_j}{\sum_{k=1}^{m} I_k}$$

Where:

- W_j = weight of constraint j
- m = total number of constraints

NOTE: The sum of all weights must equal **1**:

$$\sum_{j=1}^{m} W_j = 1$$

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5. Weighted Score per Constraint
Each normalized value is multiplied by its corresponding weight:

$$S_{ij} = N_{ij} \times W_j$$

Where:

- S_{ij} = weighted score of design i for constraint j

6. Total Design Score
The **overall score** for each design is computed by summing all weighted scores:

$$T_i = \sum_{j=1}^{m} S_{ij}$$

Where:

- T_i = total score of design i

7. Design Ranking Rule
The **best design** is determined using:

$$\text{Best Design} = \max(T_i)$$

The design with the **highest total score** is selected as the final design.

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8. Sensitivity Analysis Formulas
Sensitivity analysis tests whether the final design remains optimal under **different importance distributions**.
8.1 Modified Weight Calculation

For each iteration r:

$$W_j^{(r)} = \frac{I_j^{(r)}}{\sum_{k=1}^{m} I_k^{(r)}}$$

8.2 Recalculated Total Score

$$T_i^{(r)} = \sum_{j=1}^{m} \left(N_{ij} \times W_j^{(r)} \right)$$

8.3 Robustness Criterion
If the same design satisfies:

$$\arg\max_i \left(T_i^{(r)}\right) \quad \forall r$$

then the design is considered **robust** and **insensitive to weight variation**.

9. Radar Chart Data Preparation (Optional Visualization)
Normalized values are directly plotted:

$$\text{Radar Axis Values} = N_{ij}$$

Each axis corresponds to a constraint, and each line corresponds to a design.

Weighted Scoring Equations
What to Write:

- Explain that normalized values are multiplied by importance percentages
- Present a general scoring equation for each design
- Define every variable used in the equations

What to Include:

- Separate example equations for each design
- Explanation that total scores determine ranking

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3.2.1 Tradeoff 1: [Constraint Name]
Each constraint must be discussed **individually** and follow the **same structure**.
What to Write:

- Define the constraint
- Explain why it matters to the client or system
- Identify whether it is a **minimization or maximization** criterion

What to Include:

- A table listing raw values of the constraint for all designs
- Brief interpretation of the table

3.2.1.1 Design 1: Normalization of [Constraint]

3.2.1.2 Design 2: Normalization of [Constraint]

3.2.1.3 Design 3: Normalization of [Constraint]
For **each design**, include:

- Statement of raw value
- Identification of min and max values
- Step-by-step substitution into the correct equation
- Final normalized value

NOTE: Show full computations

3.2.2 Tradeoff 2: [Constraint Name]
Repeat the **exact same structure** used in Tradeoff 1:

- Definition
- Raw values table
- Normalization for each design

3.2.3 Tradeoff 3: [Constraint Name]
Follow the same format and depth.

3.2.4 Tradeoff 4: [Constraint Name]
Follow the same format and depth.

3.2.5 Tradeoff 5: [Constraint Name]
Follow the same format and depth. Consistency is mandatory for credibility.

Summary of Normalized Values
To consolidate all normalized scores into one table.
What to Include:

- A table summarizing normalized values for all designs across all constraints
- Brief explanation of which designs scored highest per constraint

Designers' Raw Ranking for the Designs
To apply **importance weighting** to constraints.
What to Write:

- Explain how importance values are assigned (scale of 0–10)
- Justify why some constraints are more important than others

What to Include:

- A table showing:
    - Constraint
    - Importance score
    - Percentage weight
    - Ability-to-satisfy scores
    - Total score per design

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3.3 Influence of the Design Tradeoff in the Final Design
To explain how tradeoffs affected final decision-making.
What to Write:

- Restate how constraints guided selection
- Refer to computed total scores
- Identify the winning design based on objective evaluation

Include a final results table summarizing design scores.

3.4 Sensitivity Analysis
To test whether the winning design remains consistent under different importance rankings.
What to Write:

- Define sensitivity analysis
- Explain why multiple importance combinations are tested

3.4.1–3.4.x Sample Iterations
For each iteration:

- State the importance ranking used
- Identify which constraint has highest and lowest importance
- Present a results table

- State the winning design

Use multiple iterations to demonstrate robustness.

3.4.x Conclusion of Sensitivity Analysis

What to Write:

- Summarize results across all iterations
- State whether the same design consistently wins
- Mention use of graphical tools (e.g., radar chart)

Visualization (Radar Chart)

What to Include:

- Description of the chart
- Explanation of axes and lines
- Interpretation showing why the winning design is superior

NOTE: WRITING STYLE REQUIREMENTS

- Formal, analytical tone
- Objective, computation-based explanations
- No new data introduced
- Clear references to tables, equations, and figures

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CHAPTER 4: FINAL DESIGN

Chapter 4 presents the **final selected design**, explains **how it works in actual implementation**, and proves that it **meets the objectives and requirements stated in Chapter 1**. This chapter also documents the **testing procedures, evaluation methods, and results** to validate the system's performance, accuracy, and reliability.

This chapter is **implementation- and results-oriented**, not conceptual.

Chapter Introduction

What to Write:

- State that the final design was selected based on Chapter 3 tradeoff and sensitivity analyses
- Identify which design emerged as the best solution
- Explain that the chapter will:
  - Describe the final hardware/system configuration
  - Present the final software implementation
  - Discuss testing procedures and evaluation
  - Present test and evaluation results

4.1 Final Design

To present a **complete description of the winning design**.

What to Write:

- Name the final design
- State why this design was selected
- Give an overview of the system and its main components
- Refer to a figure showing the overall final design

What to Include:

- Labeled figure of the final system
- Brief description of each visible major component
- Explanation of how components work together

Avoid discussing test results here.

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Views of the Final Design (Figures)

What to Write:

For each system view (e.g., front, side, internal, or component views):

- Identify the purpose of the view
- Describe key components visible in that view
- Explain the role of each component in system operation

What to Include:

- Multiple figures showing:
  - External structure
  - Internal layout
  - Control or enclosure units
- Clear figure numbering and captions

Internal Components Description

What to Write:

- Describe internal components housed inside enclosures or control units
- Explain the function of each component
- Mention compliance with safety and labeling standards if applicable

This section demonstrates **proper system integration and safety awareness**.

4.1.1 Software Design

To present the **final implemented software**, focusing on user interaction and system behavior.

What to Write:

- State that this subsection describes the final software interface
- Mention the platform used (general, not code-level)

- Explain that figures show the actual running system

What to Include:

Describe each software window or screen **one by one**, following the order in which users encounter them:

- Login or authentication screen (if applicable)
- Instruction or help screen
- Main interface or dashboard
- About or information page
- Settings or configuration window
- Developer or diagnostic mode (if applicable)
- Error and warning messages

For each window:

- Explain its purpose
- Describe available buttons or controls
- Explain how the user interacts with it
- Reference the corresponding figure

Do **not** include source code.

4.2 Test Procedure and Evaluation

To explain **how the system was tested** and **how its performance was evaluated**.

What to Write:

- State that test procedures are based on the objectives in Chapter 1
- Mention applicable engineering or research standards

4.2.1 Test Procedures

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What to Write:

- Present a **step-by-step procedure** for operating and testing the system
- Write procedures in **numbered format**
- Ensure steps are clear, repeatable, and chronological

What to Include:

- Power-up steps
- System initialization
- User interaction steps
- Operation flow
- Shutdown or reset procedures

This section must be detailed enough that **another researcher could replicate the test**.

Test Scenarios

What to Write:

- Identify all test cases used to validate the system
- Explain the purpose of each test case

What to Include:

Each test case description must specify:

- What is being tested (functionality or accuracy)
- Test conditions
- Number of trials conducted
- Criteria for success or failure

4.2.2 Test Evaluation

To explain **how test results were evaluated and judged**.

What to Write:

- Identify the standard used for accuracy and precision

- State acceptable margin of error or performance threshold

What to Include:

- Accuracy formula
- Average accuracy formula
- Definition of all variables in the equations
- Explanation of why the standard is appropriate

4.3 Test and Evaluation Results

To present **actual test results and findings**.

What to Write:

- Brief overview of where and how testing was conducted
- Statement that results are based on defined procedures

4.3.1 Test Results

Each test case must be presented **individually**.

What to Include for Each Case:

- Description of the test case
- Table of results showing trial data
- Computed accuracy values
- Remarks column indicating accuracy or inaccuracy
- Brief interpretation of results

Tables must include:

- Trial number
- Expected output
- Actual/system output
- Accuracy or remarks

Only partial tables may be shown in the chapter; remaining data should be referenced in appendices.

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4.3.2 Evaluation Results

To present **client or evaluator feedback**.

What to Write:

- Identify the evaluator
- Explain the evaluation method used
- Describe evaluation criteria

What to Include:

- Evaluation rubric table
- Criteria such as:
    - Ease of use
    - Performance
    - Reliability
    - Efficiency
    - Safety
    - Quality of fabrication or implementation
- Summary of ratings
- Identified strengths and weaknesses

4.4 Conclusion

What to Write:

- Restate the project objectives
- Summarize how the final design met those objectives
- Reference successful test and evaluation outcomes

- State the overall success of the project

Avoid introducing new data.

4.5 Impact of the Design to the Community

To explain the **broader significance** of the project.

What to Write:

- Identify the problem addressed by the design
- Explain how the design improves existing practices
- Discuss benefits to users, organizations, or society
- Relate the project to sustainability or development goals if applicable

This section demonstrates **social relevance and real-world value**.

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CHAPTER 5: TECHNOPRENEURSHIP REPORT

This guide explains **how to write Chapter 5 (Technopreneurship Report)** strictly following the **structure, sequence, depth, and presentation style** of the attached document. Although the reference document discusses a specific product and company, this guide is written for a **general study**. Replace all names, figures, prices, and locations with details relevant to your own project.

Chapter 5 translates the **technical project developed in Chapters 1–4** into a **business and commercialization perspective**. It explains how the project can function as a product or service, how it will be marketed, how value is created, and how it can generate revenue while providing social impact.
This chapter is **descriptive and strategic**, not technical.

Chapter Introduction

What to Write:

- State that the chapter presents the **technopreneurship aspect** of the project
- Mention that it discusses:
  - The business plan
  - The product or service offering
  - Marketing plans and strategies
  - The business model
  - Intellectual property considerations

Keep the introduction concise and business-oriented.

5.1 Business Plan

To present an overview of the proposed business built around the developed system or product.

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5.1.1 Executive Summary

What to Write:

- Provide a **clear, non-technical summary** of the product or system
- Identify the **problem** being addressed
- Explain **how the product solves the problem**
- Mention:
  - Target users or customers
  - When and why the product was developed
  - The key benefit or innovation of the product

This section should be understandable to **non-technical readers**.

5.1.2 General Company Description

What to Write:

- State the **company name** and describe its identity
- Explain the **meaning of the company name or logo** (symbolism or concept)
- Describe:
  - Nature of the business (e.g., technology, healthcare, automation)
  - Company location
  - Year of establishment
  - Background of the founders (general expertise only)

What to Include:

- Figure of the company logo
- Short narrative explaining company vision and philosophy

Avoid resumes or overly personal details.

5.1.3 Products and Services
What to Write:

- Describe the **main product or service** offered
- Explain what the product does in practical terms
- State how it benefits the customer
- Clarify whether it is:
    - A physical product
    - A software solution
    - A system or service

Focus on **customer value**, not technical design.

5.1.4 Marketing Plan
What to Write:

- Explain the **current development stage** of the product (prototype, testing, ready for market)
- Describe how customer feedback was or will be gathered
- Explain how feedback influences product improvement
- Identify the **initial target market**
- Mention strategies for funding or partnerships

This section explains **how the company plans to enter the market**.

5.1.5 Marketing Strategy
To explain **how customers will be reached and persuaded**.
This section is divided into the **4Ps of Marketing**.

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5.1.5.1 Product
What to Write:

- Describe the product from the customer's point of view
- Explain what makes it useful or innovative
- Identify the primary users

5.1.5.2 Price
What to Write:

- State the **estimated cost of production**
- Explain how the selling price was determined
- Mention:
    - Variable costs
    - Fixed costs
    - Expected profit margin

Exact values may be estimated but must be reasonable.

5.1.5.3 Place / Location
What to Write:

- Identify where the product will initially be sold
- Explain distribution plans
- Mention future expansion plans if applicable

5.1.5.4 Promotion
What to Write:

- Describe promotional methods such as:

    - Social media
    - Partnerships
    - Referrals
    - Demonstrations or campaigns
- Explain how awareness will be created among target customers

5.2 Business Model

To present the **overall framework** of how the business creates, delivers, and captures value.

What to Write:

- Introduce the business model as a strategic tool
- State that the business model is summarized in a table

What to Include:

A **Business Model Table** containing the following elements:

- Key Partners
- Key Activities
- Key Resources
- Value Proposition
- Customer Relationships
- Channels
- Customer Segments
- Cost Structure
- Revenue Streams
- Social Impact

Each entry should be brief but specific.

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5.3 Intellectual Property (IP) Reports

To show awareness of **existing technologies and legal considerations**.

What to Write:

- Explain the importance of intellectual property protection
- Describe what patent searching is
- State that patent databases were reviewed
- Mention the use of:
    - Keywords
    - Patent classifications
    - Related technologies

What to Include:

- Statement that full patent search results are provided in the appendices
- Assurance that the proposed product does not directly copy existing patents

Do **not** include full patent listings in this chapter.